

MACHINE LEARNING CAPSTONE · RESEARCH PAPER

Refresh Opportunity Scoring for Search Content

A practical study of whether observable search-performance and content signals can help a content team decide which pages deserve review first.

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Window: April–June 2026

ABSTRACT

Content teams rarely struggle to find pages; they struggle to decide which pages deserve attention first. This study examines whether a small set of historical search-performance and content signals can support that decision without relying on a complicated model or making claims about Google's internal ranking system. Using the FlyRank internship warehouse, the analysis constructs a client-content comparison set for April through June 2026 and defines an observed decline outcome as a June impression level at least 20% below April. A transparent rule-based score is evaluated against Logistic Regression and Random Forest using a client-grouped holdout and Precision@50. The rule baseline reaches a measured Precision@50 of 0.80, compared with 0.68 for Random Forest and 0.56 for Logistic Regression. In this experiment, the simpler rule is the strongest measured method for prioritizing the first 50 review candidates.

01 • INTRODUCTION / PROBLEM STATEMENT

The problem is prioritization, not automation for its own sake

A content library can contain thousands of pages, while the team responsible for keeping that library useful has a much smaller review budget. A good search-intelligence workflow therefore does not need to decide everything automatically. It needs to make the first part of the queue more sensible.

The research question is: **Which published content items should be prioritized for refresh review based on observed search-performance signals and content characteristics?**

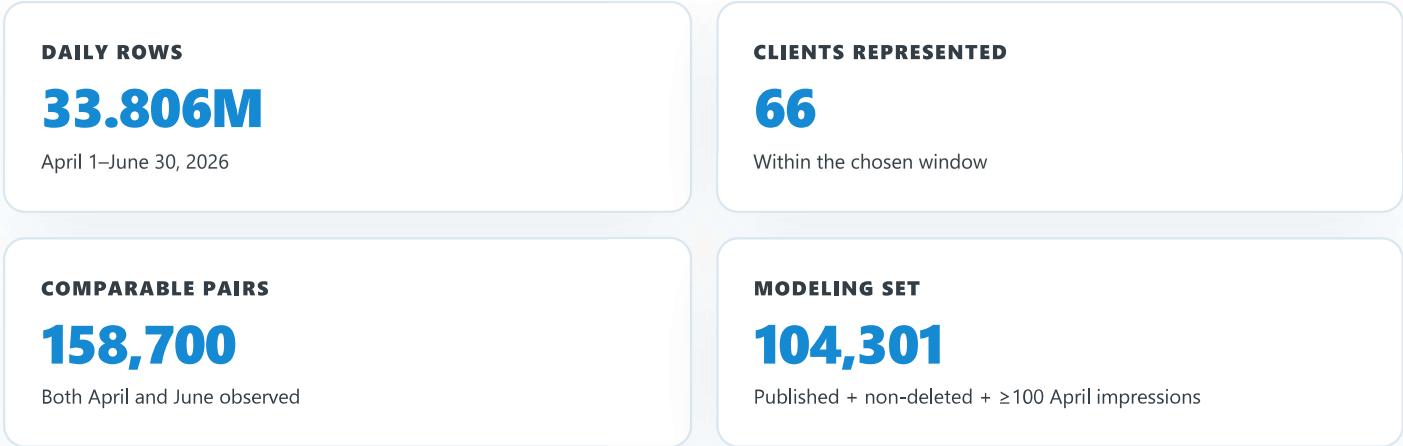
The intended user is an editorial or SEO team deciding what to investigate next. The output is a ranked review queue. It is deliberately narrower than an automated rewrite system: a high score identifies an item for earlier attention, while a human decides whether the correct action is a refresh, a metadata review, monitoring, consolidation, or no action.

Working principle: improve the quality of the review queue before adding more automation.

02 • DATA

What was measured and how the population was defined

The study uses the FlyRank internship warehouse release **v20260703**. The primary performance source is `fact_content_daily_performance`; the contextual source is `dim_content`. The daily performance table provides search-performance measurements, while the content dimension provides information about the item itself.



Observation window

April is the historical reference period and June is the later observed outcome period. Requiring both periods helps avoid treating missing history as evidence of decline. A separate June-only development sample was used while constructing the remote SQL logic so that large warehouse scans were not repeatedly performed during iteration.

Grain and duplicate check

The daily source was checked at the **report_date** × **client** × **content** grain. The June development sample contained 11,694,072 rows. A grain probe identified 6,390 additional duplicate rows. The duplicate records inspected were exact matches across their available fields, so they were removed rather than counted as separate performance observations.

DECISION	OBSERVED RESULT	WHY IT MATTERS
Compare April with June	158,700 client-content pairs	Provides a direct two-period observation
Published and non-deleted only	104,301 final modeling records	Keeps the review queue focused on live content
Minimum 100 April impressions	Applied to modeling population	Reduces extreme percentage changes caused by tiny denominators
Identifiers excluded from prediction	Used only for joins, grouping and splitting	Prevents identity from becoming a predictive shortcut

Coverage caveat. Client history is uneven across the warehouse. The analysis therefore works with records where the required GSC observations are available instead of assuming identical history for every client.

03 • METHODOLOGY

Historical inputs → controlled evaluation → review queue

01

Historical inputs

April search performance and content context.

02

Observed outcome

June impressions at least 20% below April.

03

Grouped holdout

Clients stay entirely in train or holdout.

04

Comparison

Rule, Logistic Regression, Random Forest.

05

Action

Rank items that deserve review first.

Feature design

The feature set is historical rather than outcome-aware. It uses April impressions, clicks, CTR and average position, along with search volume, competition, backlinks, character count, word count, content age and days since update. Where metadata was missing, missingness was represented explicitly before imputation instead of silently converting absence to zero.

HISTORICAL INPUTS

April impressions · April clicks · April CTR · April average position · search volume · competition · backlinks · character count · word count · content age · days since update

KEPT OUT OF THE MODEL

June performance · April-to-June change · decline label · client/content identifiers as predictive features

Observed outcome

A record is labeled `is_declining = 1` when June impressions are at least 20% below April impressions. In the final 104,301-record modeling population, 77,431 records were labeled declining and 26,870 were not.

Scope of the target. “Declining” is a measured outcome in the warehouse. It is not a direct observation of Google's ranking state and does not explain why the movement occurred.

Baseline

The baseline is deliberately transparent. It combines three signals that a reviewer can understand without a model explanation layer: historical impression opportunity, low-CTR opportunity, and freshness opportunity. Their weights are 40%, 35%, and 25% respectively. The point of the baseline is to establish what a simple rule can accomplish before introducing model complexity.

Validation design

The final comparison uses a client-grouped holdout. This avoids placing records from the same client on both sides of the evaluation, which could otherwise produce an overly optimistic estimate

when client-specific patterns are shared.

SPLIT	RECORDS	CLIENTS
Training	99,796	35
Holdout	4,505	9
Measured client overlap	0	—

Evaluation metric

The main metric is **Precision@50**: the fraction of the first 50 ranked records that are actually labeled declining in the holdout. This is intentionally aligned with the practical question of which small group of pages deserves the team's first attention.

04 · RESULTS

The simplest approach was strongest in this experiment

The baseline and both machine-learning models were evaluated on the same holdout. That gives the comparison a single reference point: the same unseen records, the same top-50 decision, and the same target definition.

Rule baseline	0.80
Random Forest	0.68
Logistic Regression	0.56

APPROACH	PRECISION@50	DECLINING IN TOP 50	DIFFERENCE VS BASELINE
Rule baseline	0.80	40 / 50	—
Random Forest	0.68	34 / 50	-0.12
Logistic Regression	0.56	28 / 50	-0.24

Main finding. The transparent rule-based ranking achieved the best measured top-50 precision on the selected client-grouped holdout. In other words, the tested ML models did not provide enough measured lift to justify their added complexity in this setup.

This should not be read as a general claim that machine learning is inferior to rules. It is a narrower finding: with this feature set, outcome definition, baseline, model configuration, time window and holdout design, the baseline was the strongest of the tested approaches.

The useful lesson is about evidence. Complexity should earn its place in the workflow by improving a measured decision, not by being present simply because the project is called machine learning.

05 • FINDINGS

What a search-content team can take from the result

1. Strong declines are useful for triage

Content with meaningful historical visibility and substantial observed decline is a sensible place to start a human investigation.

2. Explainable rankings have operational value

A reviewer can understand why a page appears high in the queue without opening a separate model-explanation interface.

3. Top-of-queue quality matches the real decision

Precision@50 is intentionally focused on the first batch of work rather than treating every row in a very large dataset as equally important.

4. Ranking is not diagnosis

A measured decline can have many causes. The score tells the reviewer where to look; it does not prove why a page moved.

06 • RANKED RECOMMENDATIONS

Turn the score into a human review workflow

Because the baseline was the best measured approach, the final queue uses it as the prioritization engine. The public output remains anonymized and does not publish client or content identifiers.

1. **Review high-priority candidates first.** Confirm that the observed decline is meaningful and that the content still matters strategically.
2. **Inspect the reason behind the score.** A strong decline, weak historical CTR and older content imply different review questions.
3. **Check the surrounding search context.** Consider intent, competition, technical health and current content quality before editing.
4. **Keep monitoring as a valid outcome.** Movement is not an automatic instruction to rewrite.

REASON CODES

STRONG_DECLINE

LOW_CTR_OPPORTUNITY

OLDER_CONTENT

MIXED_SIGNAL

HUMAN REVIEW CHECKLIST

Strategic importance · current demand · search intent · technical health · external events · competing pages · freshness · refresh versus monitor versus consolidate

07 • LIMITATIONS & HONEST FRAMING

Keep the claim smaller than the evidence

The study measures relationships in an observational dataset. It does not establish a causal mechanism for search movement. The decline label is a practical proxy, not a direct measurement of a search engine's internal state.

LIMITATION	WHAT IT PREVENTS US FROM CLAIMING
Observed impression decline is a proxy	We did not measure Google's internal ranking system.
One selected time window	The result is not guaranteed to repeat in every future period.
One grouped holdout	The measured Precision@50 is not a guarantee of future performance.
Uneven metadata coverage	Every contextual variable cannot be treated as equally reliable.
Observational design	We cannot claim that refreshing a page causes recovery.

Preferred framing: observed, measured, associated with, directional, decision-support, candidate for review.

08 · REPRODUCIBILITY

What a reviewer can inspect

The repository contains the analysis notebook, generated outputs, deployed paper and the exact location of the paper. The page is designed to be understandable without requiring the reviewer to run the full warehouse query.

ARTIFACT	ROLE
<code>work/notebooks/capstone.ipynb</code>	Capstone analysis and supporting code

ARTIFACT	ROLE
work/outputs/capstone_model_comparison.csv	Model-versus-baseline result table
work/outputs/capstone_ranked_recommendations.csv	Ranked public-safe recommendation output
submission/paper_url.txt	Direct URL used to locate the deployed paper

Public-safe publication

Client names, raw domains, private queries, credentials and raw private exports are intentionally excluded. Identifiers are used only for grouping, joins and validation.

Acknowledgments & data credit

Built on the **FlyRank ML Internship dataset**. [FlyRank](#).

Repository: github.com/Muneebshah1192/flyrank-ml-internship

AT A GLANCE

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Best measured approach
Rule baseline

Precision@50
0.80

Holdout client overlap
0

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